

SUMED CHALAKH

Data Scientist | AI/ML Engineer | Nagpur, Maharashtra
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PROFESSIONAL SUMMARY

Recent B.Tech graduate with hands-on experience in machine learning, generative AI, and data pipeline development. Skilled in Python, PySpark, and SQL, with experience building predictive models, multi-agent RAG systems, and deploying real-time inference APIs across datasets of 2M+ records. Familiar with SHAP-based model explainability, LLM orchestration (LangGraph), statistical analysis, and feature engineering. Eager to contribute to data-driven teams while growing expertise in machine learning and scalable AI systems.

SKILLS

Programming Languages: Python, SQL, JavaScript

ML Frameworks & Libraries: Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, Random Forest, Gradient Boosting

Generative AI & LLM Engineering: LangGraph, RAG (Retrieval-Augmented Generation), Multi-Agent Orchestration, Hybrid Search & Reranking, Gemini API

Data Engineering & Big Data: PySpark, Pandas, NumPy, Data Cleaning, Feature Engineering

Data Visualization & BI: Power BI, Advanced MS Excel, DAX, Streamlit, Matplotlib, Seaborn

Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions (CI/CD), Kubernetes, Prometheus

Databases & Tools: Flask, REST API, Jupyter Notebook, Google Colab, FastAPI, PostgreSQL, Qdrant (Vector DB), Redis

Specialized Skills: SHAP Explainability, Hyperparameter Tuning (GridSearchCV), Statistical Analysis, A/B Testing, Model Evaluation

PROFESSIONAL EXPERIENCE

Data Science Intern

Zaalima Development Pvt. Ltd. (Remote)

Jan 2026 – Apr 2026

- Led end-to-end ML workflows processing 2M+ records; designed 12+ feature engineering transformations using Python, SQL (JOINS, CTEs), and Pandas; delivered EDA reports identifying 15+ business insights
- Trained and tuned predictive models (Linear Regression, Decision Trees, XGBoost) using GridSearchCV; improved MAE from **15.2** → **8.7** on **200K** test set (43% reduction)
- Built automated ETL pipeline in Pandas/NumPy processing **1M+ daily records**; reduced manual data prep time from **6 hours to 30 minutes** (92% efficiency gain)

AI-ML Virtual Intern

Google (via AICTE & EduSkills Foundation)

Jun 2025 – Sep 2025

- Engineered CNN (5-layer) and RNN (LSTM) architectures using TensorFlow/Keras; CNN **achieved 94% accuracy** on image classification; RNN improved sequence recall to **89%**
- Optimized deep learning models via hyperparameter tuning; **improved F1-score** from **0.82** → **0.91** (precision 88% → 94%, recall 85% → 89%)

PROJECTS

NexusAI — Multi-Agent RAG Enterprise Knowledge Platform

Python, FastAPI, Next.js 15, PostgreSQL, Qdrant, AWS S3, Gemini 2.5 Flash, Docker

- Built a full RAG pipeline (upload → parse → chunk → embed → hybrid retrieve → rerank → cite) processing PDF/DOCX/TXT/CSV into **512-token chunks (64-token overlap)**, combining dense + BM25 search via Reciprocal Rank Fusion with cross-encoder reranking and citation-grounded generation
- Orchestrated a **4-agent system** (RAG, document-scoped, SQL, web) using LangGraph with LLM-based

conditional routing and parallel multi-agent synthesis; SQL agent uses intent-classification + parameterized dispatch instead of LLM-generated SQL to eliminate injection risk

- Implemented JWT auth, Redis-backed conversational memory, and **multi-tenant isolation** enforced at the query level across S3, PostgreSQL, and Qdrant
- Shipped via hardened multi-stage Docker builds and GitHub Actions CI/CD to AWS, with Prometheus/Grafana/Loki observability and Kubernetes manifests for horizontal scaling — built as a public, day-by-day **28-day** roadmap with a commit + design doc per day

GitHub: github.com/sumedChalakh/nexusai-enterprise-platform

FactoryGuard AI — Predictive Maintenance API

Python, Flask, SHAP, Docker, GitHub Actions, Scikit-learn, Prometheus

- Built production-ready ML REST API scoring equipment failure risk in real-time; achieved **18.88ms** average latency (p95: 32ms) with **0% error rate** across **50+ benchmark requests**
- Implemented SHAP-based /explain endpoint enabling maintenance engineers to identify root-cause sensor anomalies; added Prometheus metrics monitoring request latency, error rates, and predictions
- Containerized application using Docker; built automated testing suite (9 pytest cases); implemented GitHub Actions **CI/CD pipeline** for automated testing on every push

GitHub: github.com/sumedChalakh/factoryguard-ai

SparkScale Churn — Distributed Churn Prediction

Python, PySpark 3.5.3, MLlib, Docker, scikit-learn, Pandas, matplotlib

- Built distributed ML pipeline processing **2M IBM Telco Churn** records on Docker Spark cluster (1 master + 2 workers); engineered 5 domain-specific features via Spark SQL
- Trained Logistic Regression, Random Forest, and Gradient Boosting models via MLlib CrossValidator; GBT achieved **AUC 0.8759, F1 0.815**, outperforming LR baseline by **3.3%**
- Applied threshold tuning (0.50 → 0.30); boosted churn recall from **57% to 78.5%** (47% reduction in false negatives); delivered batch scoring job with ROC/PR curves and feature importance

GitHub: github.com/sumedChalakh/sparkscale-churn

EDUCATION

B.Tech in Civil Engineering

JD College of Engineering, Nagpur | Graduated: May 2026 | CGPA: 8.4/10

Relevant focus: Statistical Analysis, Quantitative Methods, Data-Driven Infrastructure Planning

CERTIFICATIONS

AWS Cloud Computing Essentials — Amazon Web Services (Completed Mar 2026)

Google AI-ML Virtual Internship — AICTE & EduSkills Foundation (Completed Sep 2025)

Python Full Stack Development — EduSkills Academy (Completed 2025)

Power BI Professional Analytics — Skillcourse (Completed 2026)

Data Science & Machine Learning Bootcamp — Skillians (In Progress, Expected Jul 2026)